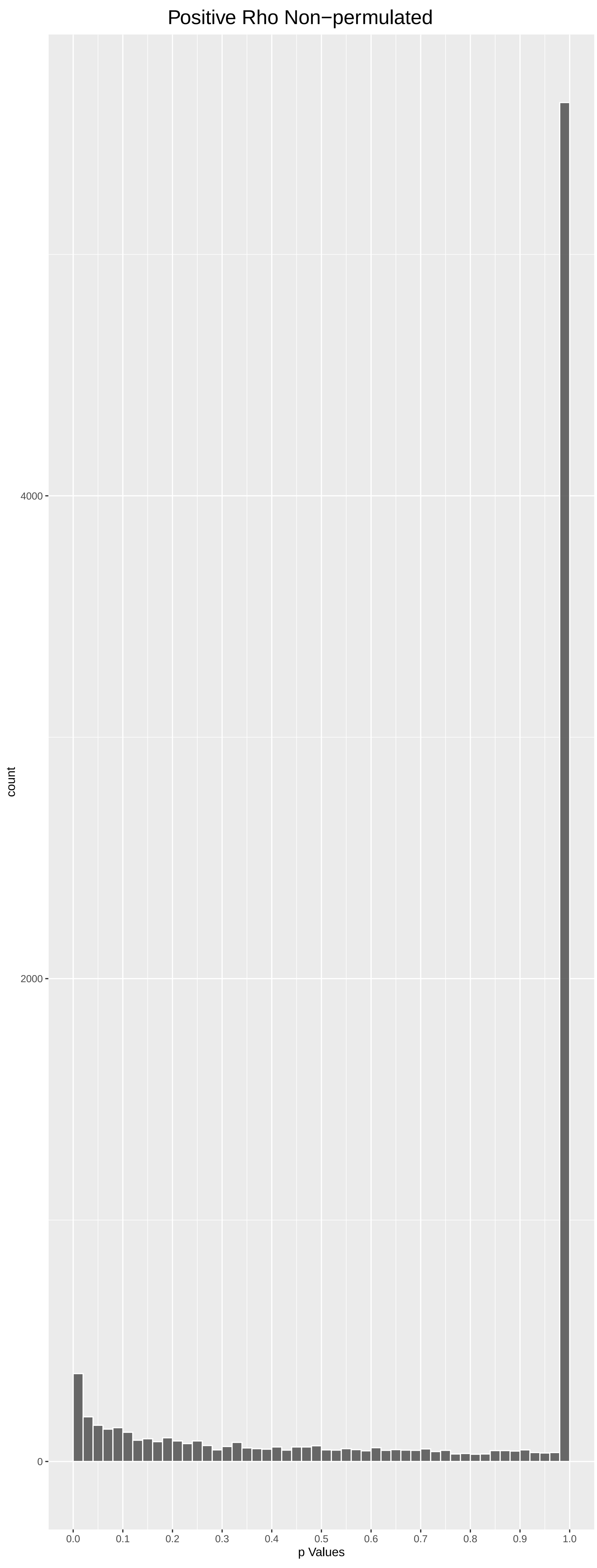
Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
SPINT4	5.038106	2.820964e−06	1.420e−02	1.000e+00
CATSPERE	5.228462	1.025557e−06	1.420e−02	1.000e+00
NDUFS5	5.042299	2.759832e−06	1.420e−02	1.000e+00
CATSPERB	4.924687	5.069737e−06	1.914e−02	1.000e+00
SCFD1	4.876634	6.474687e−06	1.956e−02	1.000e+00
ATP2B4	4.660961	1.888415e−05	2.852e−02	1.000e+00
GPR143	−4.673620	1.775617e−05	2.852e−02	1.000e+00
RANBP17	4.731777	1.335380e−05	2.852e−02	1.000e+00
PCDH15	−4.699912	1.561641e−05	2.852e−02	1.000e+00
DYNC2H1	−4.735616	1.310343e−05	2.852e−02	1.000e+00
MRGPRF	−4.620083	2.301517e−05	3.160e−02	1.000e+00
PKD1L1	−4.591508	2.640328e−05	3.323e−02	1.000e+00
FBXW12	4.508214	3.922537e−05	4.557e−02	1.000e+00
CEMIP	−4.411816	6.150437e−05	4.673e−02	1.000e+00
TMEM126A	4.411631	6.155678e−05	4.673e−02	1.000e+00
METAP1D	4.483983	4.395747e−05	4.673e−02	1.000e+00
DNMT3B	4.402768	6.412692e−05	4.673e−02	1.000e+00
DLG1	4.429718	5.661372e−05	4.673e−02	1.000e+00
R3HDM4	4.403929	6.378457e−05	4.673e−02	1.000e+00
FUBP1	4.394575	6.659368e−05	4.673e−02	1.000e+00
MYH13	−4.389821	6.806652e−05	4.673e−02	1.000e+00
TRANK1	−4.441340	5.364026e−05	4.673e−02	1.000e+00
ARHGAP36	4.337808	8.634644e−05	5.434e−02	1.000e+00
SLC22A16	4.346012	8.318117e−05	5.434e−02	1.000e+00
NTAQ1	4.310829	9.758605e−05	5.459e−02	1.000e+00

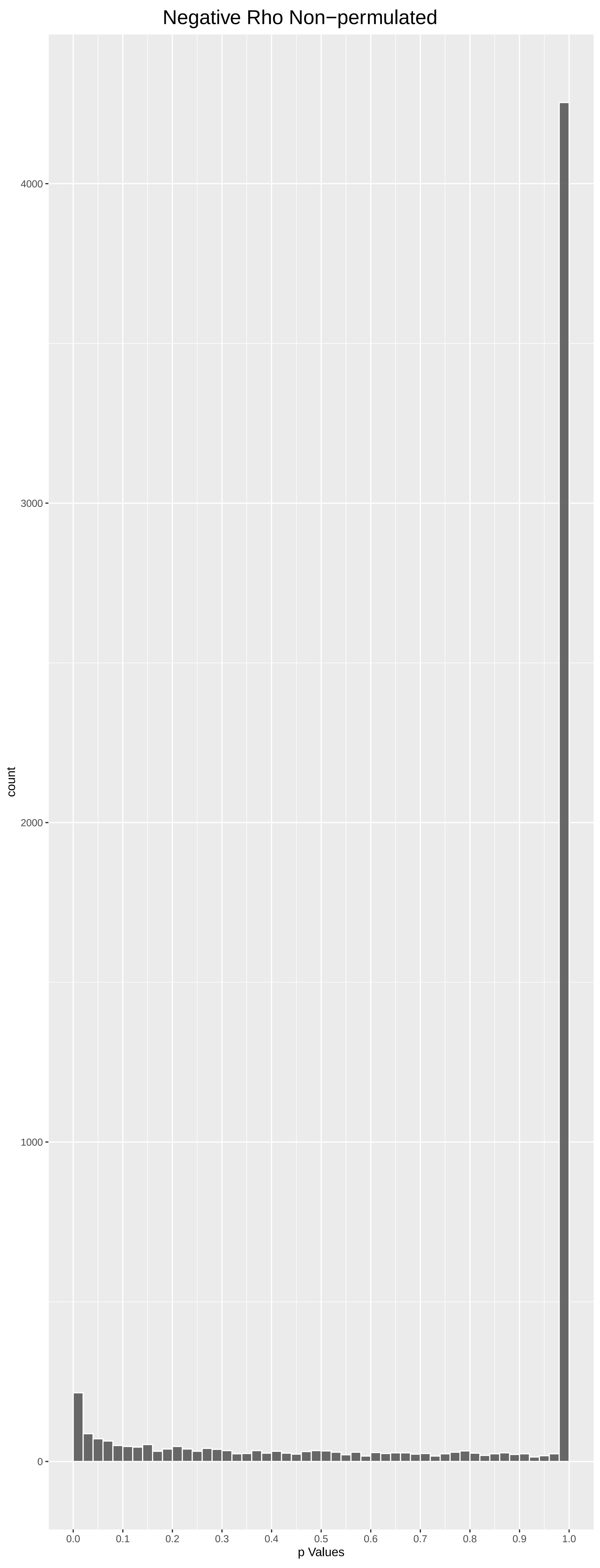
Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
M6PR	−0.64494358	1.00000000	1.000e+00	1.000e+00
FKBP4	−0.81787552	1.00000000	1.000e+00	1.000e+00
CYP26B1	0.69095307	1.00000000	1.000e+00	1.000e+00
NDUFAF7	0.66702789	1.00000000	1.000e+00	1.000e+00
FUCA2	0.39998810	1.00000000	1.000e+00	1.000e+00
DBNDD1	−0.14536960	1.00000000	1.000e+00	1.000e+00
HS3ST1	0.27155118	1.00000000	1.000e+00	1.000e+00
SEMA3F	1.70024256	0.53451186	1.000e+00	1.000e+00
CFTR	−0.10994171	1.00000000	1.000e+00	1.000e+00
CYP51A1	2.77683152	0.03293497	6.568e−01	1.000e+00
USP28	−0.92004574	1.00000000	1.000e+00	1.000e+00
SLC7A2	1.16019148	1.00000000	1.000e+00	1.000e+00
HSPB6	0.69540687	1.00000000	1.000e+00	1.000e+00
PDK4	−0.65016151	1.00000000	1.000e+00	1.000e+00
USH1C	−0.85281508	1.00000000	1.000e+00	1.000e+00
BAIAP2L1	0.15111799	1.00000000	1.000e+00	1.000e+00
TMEM132A	−2.11882836	0.20462985	1.000e+00	1.000e+00
DVL2	1.09101381	1.00000000	1.000e+00	1.000e+00
RPAP3	0.15655566	1.00000000	1.000e+00	1.000e+00
HOXA11	0.14874338	1.00000000	1.000e+00	1.000e+00
TRAPPC6A	0.46011468	1.00000000	1.000e+00	1.000e+00
SPATA20	0.05511899	1.00000000	1.000e+00	1.000e+00
CD79B	0.78751794	1.00000000	1.000e+00	1.000e+00
RHBDD2	−1.46012452	0.86553513	1.000e+00	1.000e+00
RPUSD1	0.01171201	1.00000000	1.000e+00	1.000e+00

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
SPINT4	5.038106	2.820964e−06	1.420e−02	1.000e+00
CATSPERE	5.228462	1.025557e−06	1.420e−02	1.000e+00
NDUFS5	5.042299	2.759832e−06	1.420e−02	1.000e+00
CATSPERB	4.924687	5.069737e−06	1.914e−02	1.000e+00
SCFD1	4.876634	6.474687e−06	1.956e−02	1.000e+00
ATP2B4	4.660961	1.888415e−05	2.852e−02	1.000e+00
RANBP17	4.731777	1.335380e−05	2.852e−02	1.000e+00
FBXW12	4.508214	3.922537e−05	4.557e−02	1.000e+00
TMEM126A	4.411631	6.155678e−05	4.673e−02	1.000e+00
METAP1D	4.483983	4.395747e−05	4.673e−02	1.000e+00

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
GPR143	−4.673620	1.775617e−05	2.852e−02	1.000e+00
PCDH15	−4.699912	1.561641e−05	2.852e−02	1.000e+00
DYNC2H1	−4.735616	1.310343e−05	2.852e−02	1.000e+00
MRGPRF	−4.620083	2.301517e−05	3.160e−02	1.000e+00
PKD1L1	−4.591508	2.640328e−05	3.323e−02	1.000e+00
CEMIP	−4.411816	6.150437e−05	4.673e−02	1.000e+00
MYH13	−4.389821	6.806652e−05	4.673e−02	1.000e+00
TRANK1	−4.441340	5.364026e−05	4.673e−02	1.000e+00
CEP164	−4.281717	1.112740e−04	5.795e−02	1.000e+00
FAM217B	−4.269865	1.173546e−04	5.908e−02	1.000e+00

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
'De Novo' AMP Biosynthetic Process (GO:0	−0.08255843	4	5.674e−01	9.886e−01	PFAS:111 PAICS:7003.5 ATIC:7003.5 ADSS1:7003.5 NA NA NA
'De Novo' Post−Translational Protein Fol	0.13105097	15	7.895e−02	9.886e−01	DNAJB13:142 SDF2L1:644 HSPA8:799 HSPA14:923 PTGES3:5662.5 HSPA2:5662.5 DNAJB14:5662.5
2−Oxoglutarate Metabolic Process (GO:000	0.08438378	13	2.922e−01	9.886e−01	GOT1:14 OGDHL:660 GOT2:675 IDH1:5662.5 PHYH:5662.5 ADHFE1:5662.5 GPT2:5662.5
3'−UTR−mediated mRNA Destabilization (GO	−0.02413949	11	7.816e−01	9.886e−01	DND1:1094 ZFP36:2084 TRIM71:7003.5 RC3H1:7003.5 ZFP36L1:7003.5 ZFP36L2:7003.5 MOV10:7003.5
3'−UTR−mediated mRNA Stabilization (GO:0	−0.06303912	11	4.692e−01	9.886e−01	ELAVL4:1331 YBX3:1592 MYD88:2044 ZFP36:2084 TARDBP:7003.5 MAPK14:7003.5 DAZL:7003.5
3'−Phosphoadenosine 5'−Phosphosulfate Me	0.10660456	8	2.965e−01	9.886e−01	SULT1A2:323 PAPSS2:5662.5 TPST2:5662.5 SULT2B1:5662.5 SULT1B1:5662.5 SULT1E1:5662.5 ENPP1:5662.5
5S Class rRNA Transcription By RNA Polym	−0.23755463	5	6.584e−02	9.886e−01	GTF3C5:373 GTF3C3:578 GTF3C4:1652 GTF3C2:7003.5 GTF3C6:7003.5 NA NA
7−Methylguanosine RNA Capping (GO:000945	0.26546942	2	1.935e−01	9.886e−01	RNMT:311 CMTR2:5662.5 NA NA NA NA NA
7−Methylguanosine mRNA Capping (GO:00063	0.26546942	2	1.935e−01	9.886e−01	RNMT:311 CMTR2:5662.5 NA NA NA NA NA
ADP Transport (GO:0015866)	−0.12074398	5	3.498e−01	9.886e−01	SLC25A24:813 SLC25A23:2170 SLC25A42:7003.5 SLC25A41:7003.5 SLC25A5:7003.5 NA NA
AMP Biosynthetic Process (GO:0006167)	−0.11099817	6	3.464e−01	9.886e−01	PFAS:111 APRT:1396 PAICS:7003.5 ATIC:7003.5 ADSS1:7003.5 NUDT2:7003.5 NA
AMP Metabolic Process (GO:0046033)	0.04567587	9	6.352e−01	9.886e−01	AK9:324 ADSS1:5662.5 NUDT2:5662.5 NT5C1A:5662.5 NT5E:5662.5 AMPD1:5662.5 AMPD3:5662.5
ATF6−mediated Unfolded Protein Response	0.05376563	5	6.772e−01	9.886e−01	MBTPS1:5662.5 XBP1:5662.5 ATF6B:5662.5 DDIT3:5662.5 CREBZF:5662.5 NA NA
ATP Biosynthetic Process (GO:0006754)	0.07835106	13	3.281e−01	9.886e−01	ATP5F1C:339 ATP5PD:423 NUDT2:5662.5 TGFB1:5662.5 VPS9D1:5662.5 ATP5PF:5662.5 ATP5PB:5662.5
ATP Metabolic Process (GO:0046034)	−0.02145636	25	7.105e−01	9.886e−01	PRKAG2:478 NADK:762 PARP1:1028 FIGNL1:1715 MYH8:1858 CTNS:2217 ENPP1:7003.5
ATP Synthesis Coupled Electron Transport	0.04792873	3	7.737e−01	9.886e−01	NDUFB6:717 NDUFS2:5662.5 NDUFV3:10845 NA NA NA NA
ATP Transport (GO:0015867)	−0.09707394	9	3.133e−01	9.886e−01	SLC25A24:813 CALHM1:893 SLC25A23:2170 SLC25A42:7003.5 SLC25A41:7003.5 SLC25A5:7003.5 CALHM3:7003.5
AV Node Cell Action Potential (GO:008601	−0.17229959	6	1.439e−01	9.886e−01	CACNB2:335 SCN4B:1328 CACNA1G:2205 SCN10A:7003.5 SCN5A:7003.5 TRPM4:7003.5 NA
AV Node Cell To Bundle Of His Cell Commu	0.05376016	4	7.096e−01	9.886e−01	SCN5A:5662.5 CXADR:5662.5 GJC3:5662.5 GJA5:5662.5 NA NA NA
Arp2/3 Complex−Mediated Actin Nucleation	−0.04453935	10	6.258e−01	9.886e−01	WASHC1:820 ARPC1A:977 ARPC5L:2053 WHAMM:2179 ARPC5:7003.5 ARPC3:7003.5 JMY:7003.5
B Cell Activation (GO:0042113)	0.09136123	61	1.374e−02	9.886e−01	HDAC9:43 CD86:62 FLT3:159 CASP8:207 CD40:419 CD79A:527 MEF2C:761
B Cell Activation Involved In Immune Res	0.05942331	12	4.761e−01	9.886e−01	RNF168:18 MSH2:5662.5 GPR183:5662.5 CD40LG:5662.5 CD180:5662.5 LIG4:5662.5 RNF8:5662.5
B Cell Differentiation (GO:0030183)	0.08932514	44	4.058e−02	9.886e−01	HDAC9:43 FLT3:159 CD79A:527 PTPRC:780 IL4:881 SLC25A5:5662.5 PTPN2:5662.5
B Cell Proliferation (GO:0042100)	0.16028080	15	3.166e−02	9.886e−01	CD40:419 CD79A:527 MEF2C:761 PTPRC:780 IL10:5662.5 CD70:5662.5 IFNE:5662.5
B Cell Receptor Signaling Pathway (GO:00	0.09914482	26	8.032e−02	9.886e−01	CD79A:527 CTLA4:566 MEF2C:761 PTPRC:780 NFAM1:5662.5 MS4A1:5662.5 BLNK:5662.5
BMP Signaling Pathway (GO:0030509)	−0.01336705	42	7.646e−01	9.886e−01	FAM83G:157 INHBE:622 LEFTY1:835 DDX5:960 SMAD5:983 ACVR2A:1596 LEFTY2:1837
C−terminal Protein Amino Acid Modificati	−0.10195328	8	3.180e−01	9.886e−01	GPLD1:832 AGBL5:2195 AGBL4:2243 AGTPBP1:7003.5 FOLH1:7003.5 ATG16L1:7003.5 ATG7:7003.5
C−terminal Protein Deglutamylation (GO:0	−0.13526423	4	3.488e−01	9.886e−01	AGBL5:2195 AGBL4:2243 AGTPBP1:7003.5 FOLH1:7003.5 NA NA NA
C−terminal Protein Lipidation (GO:000650	−0.10932155	3	5.120e−01	9.886e−01	GPLD1:832 ATG16L1:7003.5 ATG7:7003.5 NA NA NA NA
C4−dicarboxylate Transport (GO:0015740)	0.03951597	10	6.653e−01	9.886e−01	SLC25A12:782 SLC25A13:5662.5 SLC13A2:5662.5 SLC25A18:5662.5 SLC13A5:5662.5 SLC1A1:5662.5 UCP2:5662.5
CD4−positive, Alpha−Beta T Cell Activati	−0.04757597	5	7.126e−01	9.886e−01	NKG7:605 STOML2:7003.5 TNFSF4:7003.5 RSAD2:7003.5 HMGB1:7003.5 NA NA
CD40 Signaling Pathway (GO:0023035)	0.11701448	7	2.837e−01	9.886e−01	CD86:62 CD40LG:5662.5 ITGB1:5662.5 PHB2:5662.5 TNIP2:5662.5 ITGA5:5662.5 RNF31:5662.5
CENP−A Containing Chromatin Assembly (GO	0.04567537	5	7.236e−01	9.886e−01	MIS18A:728 CENPI:5662.5 OIP5:5662.5 NASP:5662.5 HJURP:11122 NA NA
COPI Coating Of Golgi Vesicle (GO:004820	0.05374924	2	7.923e−01	9.886e−01	GBF1:5662.5 ARFGAP3:5662.5 NA NA NA NA NA
COPI−coated Vesicle Budding (GO:0035964)	0.05374924	2	7.923e−01	9.886e−01	GBF1:5662.5 ARFGAP3:5662.5 NA NA NA NA NA
COPII Vesicle Coating (GO:0048208)	−0.04364555	14	5.719e−01	9.886e−01	KLHL12:438 TRAPPC11:1323 TRAPPC12:2026 TRAPPC3:7003.5 TRAPPC2L:7003.5 TRAPPC4:7003.5 TRAPPC1:7003.5
CRD−mediated mRNA Stabilization (GO:0070	−0.14997459	5	2.455e−01	9.886e−01	YBX1:306 IGF2BP1:841 DHX9:7003.5 PABPC1:7003.5 HNRNPD:7003.5 NA NA
DNA Alkylation (GO:0006305)	0.07697969	15	3.021e−01	9.886e−01	DNMT3B:57 BAZ2A:397 MTRR:5662.5 HELLS:5662.5 ATRX:5662.5 DNMT3A:5662.5 EHMT1:5662.5
DNA Catabolic Process (GO:0006308)	−0.04419696	17	5.282e−01	9.886e−01	DICER1:391 REXO4:645 DFFB:881 DNASE1L3:900 DNASE1L2:1106 ENDOG:7003.5 FBH1:7003.5
DNA Conformation Change (GO:0071103)	−0.04860193	9	6.137e−01	9.886e−01	HMGB3:599 TOP3A:1756 TOP2B:7003.5 HMGB1:7003.5 TOP2A:7003.5 TOP3B:7003.5 ERCC3:7003.5

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	−0.146877141	100	4.038e−07	2.615e−03	COL20A1:29 FAM217B:76 MTG2:103 HELZ2:139 CDH26:183 ZMYND8:197 SLCO4A1:251
NIKOLSKY_BREAST_CANCER_1Q21_AMPLICON	−0.237384250	27	1.972e−05	6.384e−02	ADAM15:92 PMVK:204 RORC:206 MTX1:436 TDRKH:504 ZBTB7B:522 FDPS:557
KEGG_ALLOGRAFT_REJECTION	0.230933275	21	2.496e−04	5.388e−01	CD86:62 IFNG:93 FASLG:99 IL12B:143 HLA−DOB:232 CD80:296 CD28:361
PEREZ_TP53_TARGETS	−0.036632011	812	4.740e−04	6.139e−01	GPR143:4 CEMIP:6 APC2:25 KDM2B:26 UNKL:55 SMTNL2:67 EME2:70
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	−0.056870758	333	3.910e−04	6.139e−01	KCNH2:1 CHRNB2:3 CHRNA3:45 HTR1D:78 ZBTB16:99 HTR6:125 TDRD6:155
WP_EFFECTS_OF_NITRIC_OXIDE	−0.395603951	6	7.912e−04	8.540e−01	NOS3:244 MB:354 AOX1:1500 XDH:1618 NOS1:2017 NOS2:2233 NA
CHARAFE_BREAST_CANCER_LUMINAL_VS_MESENCH	−0.053803326	319	1.018e−03	9.419e−01	EPS8L1:52 SEMA4A:104 FAM83H:117 MLPH:152 KRT19:198 CAPN8:238 GGT6:241
SHARMA_ASTROCYTOMA_WITH_NF1_SYNDROM	−0.086348307	3	6.045e−01	9.986e−01	CLEC3B:1708 ADGRV1:7003.5 SLC1A3:7003.5 NA NA NA NA
SOGA_COLORECTAL_CANCER_MYC_DN	0.038979579	56	3.134e−01	9.986e−01	ACO2:74 SULT1A2:323 SLC44A1:614 SULT1A4:684 PLPP1:865 B3GNT5:994 SLC4A4:5662.5
SOGA_COLORECTAL_CANCER_MYC_UP	0.028444909	63	4.353e−01	9.986e−01	GPI:398 UMPS:507 SLC7A6:554 POLR3K:734 DTYMK:985 PAICS:5662.5 CAD:5662.5
CORONEL_RFX7_DIRECT_TARGETS_UP	0.014331788	23	8.120e−01	9.986e−01	XRCC1:905 TSPYL2:1000 ABAT:5662.5 KLF9:5662.5 TOP2B:5662.5 ATRX:5662.5 TP53INP1:5662.5
FOROUTAN_TGFB_EMT_UP	−0.008841825	138	7.205e−01	9.986e−01	SACS:64 MMP9:328 TUFT1:407 EML1:464 ARFGAP1:582 PLEK2:594 LARP6:634
FOROUTAN_TGFB_EMT_DN	−0.060006315	81	6.224e−02	9.986e−01	CAVIN2:69 ERMP1:186 KRT19:198 SLCO4A1:251 LAMA5:295 JAG2:319 KRT15:330
FOROUTAN_PRODRANK_TGFB_EMT_UP	−0.017890028	132	4.788e−01	9.986e−01	SACS:64 MMP9:328 TUFT1:407 EML1:464 ARFGAP1:582 PLEK2:594 LARP6:634
FOROUTAN_PRODRANK_TGFB_EMT_DN	−0.052924699	64	1.435e−01	9.986e−01	CAVIN2:69 ERMP1:186 KRT19:198 SLCO4A1:251 LAMA5:295 JAG2:319 SYBU:742
FOROUTAN_INTEGRATED_TGFB_EMT_UP	0.008761891	82	7.841e−01	9.986e−01	COL1A1:148 VGLL3:427 BMPR2:512 COL6A3:646 GALNT10:676 SLC26A2:5662.5 JAG1:5662.5
FOROUTAN_INTEGRATED_TGFB_EMT_DN	−0.048390018	57	2.067e−01	9.986e−01	CAVIN2:69 ERMP1:186 KRT19:198 KRT15:330 ATP8B1:805 BIRC3:885 ELF3:896
GLASS_IGF2BP1_CLIP_TARGETS_KNOCKDOWN_DN	0.001230343	85	9.688e−01	9.986e−01	NCEH1:720 CD2AP:750 SRPK1:882 N4BP2:928 SHMT2:5662.5 HDLBP:5662.5 HNRNPF:5662.5
HOUNKPE_HOUSEKEEPING_GENES	0.001427951	694	8.993e−01	9.986e−01	RNF168:18 MBD1:58 NARS1:94 CCAR2:126 IPO8:174 PDCD7:178 MRPS30:265
IBRAHIM_NRF1_UP	0.009226787	275	6.002e−01	9.986e−01	DNMT3B:57 MBD1:58 NCKAP1:92 NARS1:94 SCPEP1:149 FUBP1:152 SUPV3L1:224
IBRAHIM_NRF2_UP	0.019770601	358	2.016e−01	9.986e−01	NCKAP1:92 NARS1:94 TYW3:131 SUPV3L1:224 ZNF189:253 MRPS30:265 CLU:279
IBRAHIM_NRF3_UP	0.053253480	2	7.942e−01	9.986e−01	HMOX1:5662.5 RPN2:5662.5 NA NA NA NA NA
IBRAHIM_NRF1_DOWN	−0.031722755	94	2.885e−01	9.986e−01	TMEM25:72 KCNH3:95 NPHP4:116 LIG1:243 PIDD1:275 PBXIP1:569 PCLO:628
IBRAHIM_NRF2_DOWN	−0.052893290	109	5.686e−02	9.986e−01	IQGAP3:34 CDK10:102 ENTR1:226 LIG1:243 PPP1R16A:266 PIDD1:275 RHPN1:492
IBRAHIM_NRF3_DOWN	0.051267514	11	5.560e−01	9.986e−01	ARHGAP23:700 SMAP1:5662.5 NOG:5662.5 PLAT:5662.5 SEMA4C:5662.5 HECTD3:5662.5 IER2:5662.5
SEAVEY_EPITHELIOID_HEMANGIOENDOTHELIOMA	0.040657707	67	2.503e−01	9.986e−01	C17orf58:205 TNFSF15:272 CDC42EP5:564 NCEH1:720 MYL2:758 PLCD3:5662.5 SERPINE1:5662.5
BENITEZ_GBM_PROTEASOME_INHIBITION_RESPON	0.020469099	31	6.934e−01	9.986e−01	ELP6:978 ASNS:5662.5 MLF2:5662.5 GPATCH2L:5662.5 PNRC1:5662.5 SOX4:5662.5 SESN2:5662.5
BLANCO_MELO_SARS_COV_1_INFECTION_MCR5_CE	0.053275038	7	6.255e−01	9.986e−01	KIF14:5662.5 CENPE:5662.5 ASPM:5662.5 ESM1:5662.5 KIF20B:5662.5 BRCA2:5662.5 SMC4:5662.5
BLANCO_MELO_SARS_COV_1_INFECTION_MCR5_CE	0.173739275	4	2.288e−01	9.986e−01	HSD17B2:52 CYTL1:446 FGFR4:5662.5 RASL12:10386 NA NA NA
BLANCO_MELO_MERS_COV_INFECTION_MCR5_CELL	−0.022826461	187	2.832e−01	9.986e−01	SUSD4:68 KCNH3:95 RBM15:184 CDKN2AIP:344 ATP2A1:595 FHDC1:650 MED26:743
BLANCO_MELO_MERS_COV_INFECTION_MCR5_CELL	0.136302480	20	3.488e−02	9.986e−01	HSD17B2:52 REX1BD:89 HSD11B1:428 CYTL1:446 DNAJC12:581 PNMA2:681 A1BG:5662.5
BLANCO_MELO_INFLUENZA_A_INFECTION_A594_C	0.056309102	26	3.205e−01	9.986e−01	CXCL8:629 PLAUR:687 EREG:736 MRPL42:819 CCN2:5662.5 FGFBP1:5662.5 ANKRD1:5662.5
BLANCO_MELO_INFLUENZA_A_INFECTION_A594_C	−0.043790195	78	1.817e−01	9.986e−01	APC2:25 OPLAH:364 OVCA2:509 TONSL:694 SPATA2L:777 TBC1D16:794 CHPF2:848
BLANCO_MELO_HUMAN_PARAINFLUENZA_VIRUS_3_	0.006366834	132	8.010e−01	9.986e−01	TNFSF10:36 MYPN:41 STAT2:129 OAS2:281 PLEKHA4:290 HCAR2:299 CXCL11:372
BLANCO_MELO_HUMAN_PARAINFLUENZA_VIRUS_3_	0.003964052	81	9.020e−01	9.986e−01	PRODH2:40 CCN5:79 SERPINA6:139 EFCAB12:225 CAMSAP3:433 CA9:467 TUBB4A:619
BLANCO_MELO_RESPIRATORY_SYNCYTIAL_VIRUS_	0.001053896	194	9.598e−01	9.986e−01	TNFSF10:36 STAT2:129 PLEKHA4:290 HCAR2:299 TNFRSF9:312 TNF:363 CXCL11:372
BLANCO_MELO_RESPIRATORY_SYNCYTIAL_VIRUS_	−0.044303487	83	1.635e−01	9.986e−01	MCF2L:317 PADI2:348 MB:354 DACT2:430 HLA−DMB:453 SYNGR4:534 HOXB13:640
BLANCO_MELO_COVID19_SARS_COV_2_INFECTION	−0.038275814	59	3.096e−01	9.986e−01	N4BP3:176 DNAH17:360 RASGRP2:389 SDR16C5:395 INHBE:622 BCL2A1:759 PCDH1:854
BLANCO_MELO_COVID19_SARS_COV_2_INFECTION	−0.086149760	51	3.344e−02	9.986e−01	KLHDC7A:195 VIL1:236 SLC25A10:324 DACT2:430 HLA−DMB:453 SYNGR4:534 RADIL:664
BLANCO_MELO_COVID19_SARS_COV_2_INFECTION	0.007984507	240	6.714e−01	9.986e−01	TNFSF10:36 HDAC9:43 STAT2:129 OAS2:281 NEDD4:300 TNFRSF9:312 TNF:363

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Lymphoma, T−Cell, Cutaneous	0.07515422	177	5.967e−04	9.712e−01	TNFSF10:36 HDAC9:43 IFNG:93 FASLG:99 FOXP3:105 STAT2:129 SCPEP1:149
Hepatitis	0.07615196	201	2.134e−04	9.712e−01	AFP:6 AKR1D1:27 TNFSF10:36 SLCO6A1:72 IFNG:93 FASLG:99 FOXP3:105
Hepatitis C	0.04794725	498	3.012e−04	9.712e−01	AFP:6 AFM:31 TNFSF10:36 HDAC9:43 SLCO6A1:72 IFNG:93 FASLG:99
Psoriasis	0.04585194	522	4.089e−04	9.712e−01	TNFSF10:36 SLCO6A1:72 NCKAP1:92 IFNG:93 LRRK2:96 FASLG:99 FOXP3:105
Scleroderma	0.09146342	119	5.908e−04	9.712e−01	ENG:3 IFNG:93 PRSS55:98 CENPC:103 COL1A1:148 DNMT1:193 RTCA:242
Tumor Immunity	0.10674431	88	5.533e−04	9.712e−01	AFP:6 CD86:62 IFNG:93 FOXP3:105 IL33:167 APC:258 CD80:296
46, XX Testicular Disorders of Sex Devel	0.05307510	3	7.502e−01	9.739e−01	NR5A1:5662.5 RSPO1:5662.5 NR0B1:5662.5 NA NA NA NA
46, XY Disorders of Sex Development	0.05481569	12	5.109e−01	9.739e−01	SRD5A2:201 NR5A1:5662.5 LHCGR:5662.5 PAICS:5662.5 MAP3K1:5662.5 HSD17B7:5662.5 HSD17B13:5662.5
46, XY female	0.13710425	10	1.333e−01	9.739e−01	SRD5A2:201 WNT4:516 NR0B1:5662.5 HSD17B3:5662.5 DMRT2:5662.5 DMRT1:5662.5 GADD45G:5662.5
5,10−Methylenetetrahydrofolate reductase	0.10658180	15	1.531e−01	9.739e−01	HPGDS:330 AKAP9:382 ABCB1:461 SLCO1B1:510 XRCC1:905 SERPINE1:5662.5 XRCC4:5662.5
Abnormalities, Drug−Induced	0.05306987	2	7.949e−01	9.739e−01	CAT:5662.5 EPHX1:5662.5 NA NA NA NA NA
Abortion, Habitual	0.05308034	4	7.131e−01	9.739e−01	MTHFR:5662.5 F2:5662.5 F5:5662.5 VEGFA:5662.5 NA NA NA
Achondrogenesis, type IB (disorder)	0.05308557	5	6.810e−01	9.739e−01	SLC26A3:5662.5 SLC26A2:5662.5 SLC26A5:5662.5 SLC26A4:5662.5 DCN:5662.5 NA NA
Acidosis, Lactic	0.03216919	125	2.157e−01	9.739e−01	ALDH7A1:119 TRMT10C:280 G6PC:344 TNF:363 TFAM:413 NDUFA11:455 DBT:613
ACROMESOMELIC DYSPLASIA, MAROTEAUX TYPE	0.05306987	2	7.949e−01	9.739e−01	IGF1:5662.5 GDF5:5662.5 NA NA NA NA NA
Acute Myeloid Leukemia, M1	0.04840585	76	1.453e−01	9.739e−01	TNFSF10:36 FLT3:159 CD9:306 MPO:391 S100A8:392 AQP9:569 GFI1:668
Acute myeloid leukemia, minimal differen	0.10829880	15	1.465e−01	9.739e−01	FLT3:159 MPO:391 ABCB1:461 NCR1:715 FAS:5662.5 FANCB:5662.5 DNMT3A:5662.5
Adenocarcinoma of lung, stage IV	−0.02552503	11	7.695e−01	9.739e−01	ROS1:1143 FBLN2:2011 CD74:2287 TP53:7003.5 RET:7003.5 EGFR:7003.5 STK11:7003.5
Adenocarcinoma, Basal Cell	0.02708323	87	3.836e−01	9.739e−01	HOXA10:169 DNMT1:193 CASP8:207 APC:258 ACE:302 TNF:363 CXCL8:629
Adenocarcinoma, intestinal type	0.05310127	8	6.030e−01	9.739e−01	TP53:5662.5 EGFR:5662.5 CDKN2A:5662.5 MDM2:5662.5 OLFM4:5662.5 PDGFRB:5662.5 POU2F1:5662.5
Adenocarcinoma, Oxyphilic	0.03644370	89	2.357e−01	9.739e−01	HOXA10:169 DNMT1:193 CASP8:207 APC:258 ACE:302 TNF:363 CXCL8:629
Adenocarcinoma, Scirrhous	0.11453712	6	3.313e−01	9.739e−01	MET:1007 MMP15:5662.5 TIMP3:5662.5 MMP14:5662.5 FUT1:5662.5 FGFR2:5662.5 NA
Adenocarcinoma, Tubular	0.03654136	91	2.293e−01	9.739e−01	HOXA10:169 DNMT1:193 CASP8:207 APC:258 ACE:302 TNF:363 CXCL8:629
Adrenal hyperplasia, bilateral	−0.14913275	2	4.651e−01	9.739e−01	KCNJ5:1895 REN:7003.5 NA NA NA NA NA
Adrenoleukodystrophy, Neonatal	−0.04385359	13	5.841e−01	9.739e−01	HSD17B4:30 PEX2:353 PEX14:429 PEX16:1156 PEX19:7003.5 PEX1:7003.5 PEX10:7003.5
Agammaglobulinemia, non−Bruton type	0.12091262	6	3.051e−01	9.739e−01	CD79A:527 CD79B:5662.5 PIK3R1:5662.5 BTK:5662.5 BLNK:5662.5 TCF3:5662.5 NA
Aganglionosis, Colonic	−0.06866325	15	3.573e−01	9.739e−01	CAVIN2:69 FH:1034 F2R:1641 EDN3:2206 TP53:7003.5 RET:7003.5 CSGALNACT2:7003.5
Aganglionosis, Rectosigmoid Colon	−0.06296223	8	5.375e−01	9.739e−01	CAVIN2:69 EDN3:2206 RET:7003.5 UTP25:7003.5 ECE1:7003.5 NRG1:7003.5 EDNRB:7003.5
Aggressive periodontitis, generalized	0.02036744	25	7.246e−01	9.739e−01	CXCR1:349 TNF:363 CXCL8:629 IL4:881 SERPINE1:5662.5 MMP12:5662.5 PLAT:5662.5
Albinism, Ocular	−0.08184242	22	1.841e−01	9.739e−01	GPR143:4 DTNBP1:60 HPS4:469 LYST:730 AP3D1:1514 SLC45A2:1726 TYRP1:7003.5
Albinism, Oculocutaneous	0.03076540	27	5.803e−01	9.739e−01	PMEL:346 CTLA4:566 CSF3:5662.5 NRAS:5662.5 KIT:5662.5 PIK3CA:5662.5 CDKN2A:5662.5
Alkalosis, Respiratory	0.14732627	4	3.075e−01	9.739e−01	ASL:904 CA5A:5662.5 CPS1:5662.5 OTC:5662.5 NA NA NA
Alopecia, Male Pattern	0.06720986	26	2.358e−01	9.739e−01	HDAC9:43 SRD5A2:201 SUPV3L1:224 WNT4:516 NFE2L2:578 MTHFR:5662.5 IGF1:5662.5
alpha−1−Antitrypsin Deficiency, Autosoma	0.08438347	18	2.153e−01	9.739e−01	ACE:302 TNF:363 NFE2L2:578 ELANE:625 A2M:920 CAT:5662.5 SFXN1:5662.5
ALPHA−THALASSEMIA/MENTAL RETARDATION SYN	0.04468955	14	5.627e−01	9.739e−01	DAXX:594 PML:5662.5 TP53:5662.5 ALK:5662.5 SMARCA1:5662.5 ATRX:5662.5 SMARCA2:5662.5
Alzheimer's Disease, Focal Onset	0.03006435	58	4.290e−01	9.739e−01	CLU:279 ACE:302 TNF:363 MPO:391 TFAM:413 CHRNA7:415 CD2AP:750
Alzheimer Disease, Early Onset	0.04317460	80	1.827e−01	9.739e−01	CLU:279 ACE:302 MEFV:305 TNF:363 MPO:391 UBQLN1:395 TFAM:413
Alzheimer Disease, Late Onset	0.02950459	185	1.684e−01	9.739e−01	TET1:24 DNMT3B:57 LRP1:75 DNMT1:193 CLU:279 ACE:302 HPGDS:330
Amaurosis congenita of Leber, type 1	0.02538800	41	5.741e−01	9.739e−01	NFE2L2:578 CXCL8:629 RDH12:725 MEF2C:761 PTPRC:780 CEP290:952 ERBB2:5662.5
Ambiguous genitalia, male	0.05775740	6	6.242e−01	9.739e−01	SRD5A2:201 WT1:5662.5 CYP11A1:5662.5 POR:5662.5 HSD3B2:5662.5 MKS1:10758 NA

KeggReactome Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_THE_CITRIC_ACID_TCA_CYCLE_AND_R	0.14843381	146	7.201e−10	1.298e−06	NDUFS5:2 NDUFS3:22 ATP5PD:29 NDUFB5:50 NDUFA12:63 NDUFA11:69 NDUFB3:71
REACTOME_SENSORY_PERCEPTION	−0.09711505	291	1.659e−08	1.496e−05	PCDH15:2 RIPOR2:80 CTBP2:96 CALHM1:108 OR10S1:109 TRPM5:140 OR6Y1:177
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	0.16102289	99	3.412e−08	1.538e−05	NDUFS5:2 NDUFS3:22 ATP5PD:29 NDUFB5:50 NDUFA12:63 NDUFA11:69 NDUFB3:71
REACTOME_OLFACTORY_SIGNALING_PATHWAY	−0.15529358	105	4.264e−08	1.538e−05	OR10S1:109 OR6Y1:177 OR13D1:285 OR52E8:323 OR6B1:355 OR10J5:357 OR51I1:398
KEGG_OXIDATIVE_PHOSPHORYLATION	0.16588790	92	4.180e−08	1.538e−05	NDUFS5:2 NDUFS3:22 ATP5PD:29 NDUFB5:50 NDUFA11:69 NDUFB3:71 NDUFB8:117
REACTOME_COMPLEX_I_BIOGENESIS	0.21983167	48	1.429e−07	4.295e−05	NDUFS5:2 NDUFS3:22 NDUFB5:50 NDUFA12:63 NDUFA11:69 NDUFB3:71 NDUFB8:117
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	0.16255479	79	6.276e−07	1.617e−04	NDUFS5:2 NDUFS3:22 NDUFB5:50 NDUFA12:63 NDUFA11:69 NDUFB3:71 LRPPRC:113
KEGG_HUNTINGTONS_DISEASE	0.12471588	128	1.220e−06	2.749e−04	NDUFS5:2 NDUFS3:22 ATP5PD:29 NDUFB5:50 NDUFB3:71 TFAM:103 NDUFB8:117
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.09675701	205	2.103e−06	4.212e−04	LTA:47 GH2:62 IFNG:73 TNFSF8:87 CXCR2:168 IL6ST:183 TNFSF10:201
KEGG_OLFACTORY_TRANSDUCTION	−0.12153772	119	5.072e−06	9.144e−04	OR10S1:109 OR6Y1:177 OR13D1:285 OR52E8:323 OR6B1:355 OR10J5:357 CAMK2B:373
REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPT	−0.07735958	294	6.232e−06	1.021e−03	GPR143:3 HTR1D:15 SSTR4:53 TACR2:170 HTR6:175 PENK:235 CCR9:259
KEGG_PARKINSONS_DISEASE	0.13810130	89	7.110e−06	1.068e−03	NDUFS5:2 NDUFS3:22 ATP5PD:29 NDUFB5:50 NDUFB3:71 NDUFB8:117 NDUFA8:314
REACTOME_SPERM_MOTILITY_AND_TAXES	0.40651278	8	6.863e−05	9.518e−03	CATSPERB:4 CATSPER3:60 CATSPER4:186 CATSPERG:203 CATSPER2:377 CATSPERD:555 KCNU1:1291
REACTOME_METABOLISM_OF_STEROID_HORMONES	0.20501480	30	1.031e−04	1.325e−02	CGA:99 HSD11B1:198 HSD17B2:460 CYP17A1:479 SRD5A3:626 HSD3B2:649 AKR1B1:731
REACTOME_AMINE_LIGAND_BINDING_RECEPTORS	−0.17474048	41	1.103e−04	1.325e−02	GPR143:3 HTR1D:15 HTR6:175 ADRA2C:387 TAAR6:425 ADRB1:878 ADRA2A:908
REACTOME_GLUCOCORTICOID_BIOSYNTHESIS	0.43134178	6	2.536e−04	2.689e−02	HSD11B1:198 CYP17A1:479 HSD3B2:649 HSD11B2:837 CYP21A2:1918 SERPINA6:1940 NA
REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE	−0.12178077	76	2.504e−04	2.689e−02	EVPL:97 KRT78:157 PPL:186 KRT80:296 DSG2:390 PKP3:442 KRT85:514
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	0.04469539	580	3.117e−04	2.810e−02	FBXW12:8 UBR2:17 COL2A1:28 HSPA5:64 KLHL5:104 S100A8:116 CD80:118
REACTOME_SIGNALING_BY_GPCR	−0.04447095	595	2.833e−04	2.810e−02	GPR143:3 HTR1D:15 SSTR4:53 PIK3R5:78 ARHGEF11:87 TACR2:170 HTR6:175
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.19733558	28	3.046e−04	2.810e−02	HSD11B1:198 AKR1D1:222 HSD17B2:460 CYP17A1:479 SRD5A3:626 HSD3B2:649 HSD17B6:830
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	−0.06502516	260	3.454e−04	2.966e−02	GRIN3B:12 HTR1D:15 SSTR4:53 TACR2:170 HTR6:175 P2RX2:205 CHRNB2:252
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.04295700	555	6.869e−04	5.170e−02	LTA:47 GH2:62 IFNG:73 TNFSF8:87 SEH1L:95 CD80:118 CD86:119
REACTOME_INTERCONVERSION_OF_NUCLEOTIDE_D	0.18586559	28	6.703e−04	5.170e−02	AK9:46 AK6:298 CTPS2:435 NUDT13:473 DTYMK:654 NME4:1275 TXN:1435
REACTOME_GPCR_LIGAND_BINDING	−0.05098919	386	6.882e−04	5.170e−02	GPR143:3 HTR1D:15 SSTR4:53 TACR2:170 HTR6:175 TAS2R41:213 PENK:235
REACTOME_ANDROGEN_BIOSYNTHESIS	0.32184595	9	8.292e−04	5.816e−02	CGA:99 CYP17A1:479 SRD5A3:626 HSD3B2:649 HSD17B3:941 LHB:1690 SRD5A2:2441
REACTOME_SIGNALING_BY_INTERLEUKINS	0.05048378	381	8.386e−04	5.816e−02	IFNG:73 CD80:118 CD86:119 PIK3R3:164 IL6ST:183 ATF1:226 CHUK:228
KEGG_ALZHEIMERS_DISEASE	0.08656805	125	8.715e−04	5.820e−02	NDUFS5:2 NDUFS3:22 ATP5PD:29 NDUFB5:50 PPP3CC:67 NDUFB3:71 NDUFB8:117
REACTOME_EPIGENETIC_REGULATION_OF_GENE_E	0.09727688	96	1.025e−03	6.440e−02	DNMT3B:13 MYO1C:51 DEK:83 TET1:102 GTF2H1:218 H3−3A:233 DNMT1:235
REACTOME_REACTIONS_SPECIFIC_TO_THE_COMPL	0.33499411	8	1.036e−03	6.440e−02	CGA:99 MGAT2:576 MAN2A1:753 LHB:1690 CHST10:1871 MAN2A2:2996 FUT8:3093
REACTOME_G_ALPHA_12_13_SIGNALLING_EVENTS	−0.11888361	62	1.232e−03	6.912e−02	ARHGEF11:87 ARHGEF15:257 OBSCN:309 TRIO:402 TIAM1:621 ARHGEF10L:622 FGD1:820
REACTOME_INTERLEUKIN_10_SIGNALING	0.15818553	35	1.215e−03	6.912e−02	CD80:118 CD86:119 CXCL8:236 CSF3:447 CXCL10:522 LIF:547 IL12B:588
REACTOME_KERATINIZATION	−0.10263306	83	1.265e−03	6.912e−02	EVPL:97 KRT78:157 PPL:186 KRT80:296 DSG2:390 PKP3:442 KRT85:514
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.03346534	867	1.182e−03	6.912e−02	IFNG:73 DEK:83 PRDX1:89 CGA:99 PRMT5:111 LRPPRC:113 TPX2:115
REACTOME_PIWI_INTERACTING_RNA_PIRNA_BIOG	−0.22454611	17	1.356e−03	7.189e−02	MOV10L1:56 TDRD12:111 TDRD6:326 TDRKH:417 PIWIL1:841 POLR2J:945 DDX4:970
REACTOME_TNFR2_NON_CANONICAL_NF_KB_PATHW	0.09950494	78	2.439e−03	1.257e−01	LTA:47 TNFSF8:87 CHUK:228 PSMB2:270 NFKB2:618 TNF:661 FASLG:849
REACTOME_NGF_STIMULATED_TRANSCRIPTION	0.15617023	31	2.641e−03	1.323e−01	ATF1:226 TPH1:678 SH3GL3:843 JUNB:854 JUND:1065 TRIB1:1249 NAB1:1373
REACTOME_METABOLIC_DISORDERS_OF_BIOLOGIC	0.15748567	30	2.855e−03	1.391e−01	CYP17A1:479 CYP26C1:717 SLC35D1:851 CYP19A1:1423 FMO3:1586 CYP27A1:1605 CYP27B1:1652
REACTOME_MINERALOCORTICOID_BIOSYNTHESIS	0.42575367	4	3.189e−03	1.513e−01	CGA:99 HSD3B2:649 LHB:1690 CYP21A2:1918 NA NA NA
REACTOME_CRISTAE_FORMATION	0.16914177	25	3.441e−03	1.591e−01	ATP5PD:29 ATP5MG:770 DMAC2L:1097 ATP5MF:1519 ATP5PB:1675 DNAJC11:2191 ATP5F1A:2403
REACTOME_HDR_THROUGH_HOMOLOGOUS_RECOMBIN	−0.10658079	62	3.769e−03	1.699e−01	POLD2:41 EME2:225 POLE:413 HUS1:428 RTEL1:487 MRE11:660 EME1:892

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0008438 abnormal cutaneous collagen	−0.0424040037	13	5.970e−01	9.965e−01	OGN:483 DSE:2107 PLOD1:2153 PLOD3:7003.5 THBS2:7003.5 DCN:7003.5 COL3A1:7003.5
MP0002909 abnormal adrenal gland	0.1283225995	16	7.597e−02	9.965e−01	LRRK2:96 CHGB:127 WNT4:516 PTPN11:806 FOXA1:5662.5 SNAPIN:5662.5 EGLN3:5662.5
MP0000609 abnormal liver physiology	−0.0019648780	319	9.069e−01	9.965e−01	CRTC2:50 PASK:121 LPIN1:124 NOS3:244 LIPE:259 ADA:301 DICER1:391
MP0001730 embryonic growth arrest	0.0278309961	137	2.669e−01	9.965e−01	NCKAP1:92 CHD8:134 COL1A1:148 DNMT1:193 ATP6V0C:337 TUBG1:357 RNF2:445
MP0001731 abnormal postnatal growth	0.0018079489	447	9.003e−01	9.965e−01	UBR2:13 HSD17B2:52 LRP4:61 FOXP3:105 COL1A1:148 TMPRSS6:176 INPPL1:179
MP0002210 abnormal sex determination	−0.0239806641	260	1.943e−01	9.965e−01	MCPH1:13 PMS2:15 HSD17B4:30 MOV10L1:57 ZBTB16:99 LPIN1:124 TDRD6:155
MP0000604 amyloidosis	0.0103305138	30	8.452e−01	9.965e−01	MSR1:56 CLU:279 RELB:378 SLC30A3:489 LDLR:746 APOE:5662.5 NCKAP1L:5662.5
MP0000598 abnormal liver morphology	0.0145926634	389	3.418e−01	9.965e−01	DNMT3B:57 LRP1:75 IFNG:93 FOXP3:105 COL1A1:148 MAN2B1:158 SC5D:184
MP0005334 abnormal fat pad	0.0125534511	121	6.373e−01	9.965e−01	TRPV1:23 AQP7:175 ACE:302 HSD11B1:428 PPARA:548 RBL1:712 LDLR:746
MP0005332 abnormal amino acid	0.0161750910	104	5.725e−01	9.965e−01	ACE:302 IKZF3:424 DBT:613 LDLR:746 CD2AP:750 ASL:904 HNF1A:957
MP0005330 cardiomyopathy	0.0259835414	55	5.072e−01	9.965e−01	CAV3:130 TFAM:413 PPARA:548 MYL2:758 LY6E:5662.5 ABCA5:5662.5 TIMP3:5662.5
MP0002269 muscular atrophy	0.0121747086	70	7.264e−01	9.965e−01	CD86:62 CAV3:130 SUPV3L1:224 DST:546 LTN1:842 HNF1A:957 CYP11A1:5662.5
MP0001765 abnormal ion homeostasis	0.0016379507	92	9.571e−01	9.965e−01	TRPV5:183 IL6ST:336 MGAT2:422 HSPA4L:460 SLC34A3:716 CYP11A1:5662.5 IFNGR1:5662.5
MP0004043 abnormal pH regulation	0.0243916706	12	7.701e−01	9.965e−01	SOD2:5662.5 SLC4A10:5662.5 SLC12A1:5662.5 SLC4A4:5662.5 SCNN1B:5662.5 RHCG:5662.5 WNK4:5662.5
MP0008961 abnormal basal metabolism	−0.0781003668	15	2.956e−01	9.965e−01	HTR6:125 PER2:290 ATE1:423 DLL1:1692 BRD2:7003.5 DECR1:7003.5 FOXA2:7003.5
MP0003195 calcinosis	0.0573732180	7	5.993e−01	9.965e−01	SOD2:5662.5 SLC12A1:5662.5 ENPP1:5662.5 CASR:5662.5 FSHR:5662.5 ABCC6:5662.5 AGXT:5662.5
MP0002837 dystrophic cardiac calcinosis	−0.1400840925	5	2.782e−01	9.965e−01	MMP9:328 SLC2A4:1305 GSTZ1:7003.5 MYBPC3:7003.5 MYH6:7003.5 NA NA
MP0005369 muscle phenotype	0.0048097541	167	8.328e−01	9.965e−01	CAV3:130 FGFR1:251 AHSG:315 RELB:378 COL1A2:595 HOXA3:670 RBL1:712
MP0009250 abnormal appendicular skeleto	0.0152429054	52	7.051e−01	9.965e−01	COL1A1:148 APC:258 HOXA3:670 ROR1:710 COMP:955 FBN1:5662.5 CDKN1A:5662.5
MP0002998 abnormal bone remodeling	−0.0165411466	91	5.887e−01	9.965e−01	NOS3:244 GPR68:289 AXIN2:460 RAG1:685 IRAK3:707 SQSTM1:803 PHEX:818
MP0002066 abnormal motor capabilities/c	0.0062435567	834	5.751e−01	9.965e−01	ATP2B4:10 TRPV1:23 HSD17B2:52 LRP4:61 CD86:62 IFNG:93 FOXP3:105
MP0005248 abnormal Harderian gland	0.1336629452	11	1.251e−01	9.965e−01	APC:258 ALDH1A1:394 RARA:5662.5 CDKN1A:5662.5 CTNNB1:5662.5 RARB:5662.5 PRLR:5662.5
MP0002419 abnormal innate immunity	0.0215962203	296	2.143e−01	9.965e−01	MSR1:56 IFNG:93 MEPE:141 IL12B:143 COL1A1:148 FLT3:159 DUSP2:160
MP0004133 heterotaxia	−0.0523492138	25	3.660e−01	9.965e−01	LAMA5:295 LEFTY1:835 VANGL1:1275 ACVR2A:1596 DLL1:1692 LEFTY2:1837 MKS1:1908
MP0002396 abnormal hematopoietic system	0.0468213058	18	4.923e−01	9.965e−01	FLT3:159 CASP8:207 APC:258 MAN2A1:5662.5 MYC:5662.5 SNAI2:5662.5 SLC19A1:5662.5
MP0001529 abnormal vocalization	−0.0730337079	14	3.447e−01	9.965e−01	WFS1:447 DHCR7:1251 TACR1:1487 CHRM2:1921 DRD2:7003.5 ACHE:7003.5 LBR:7003.5
MP0004134 abnormal chest morphology	0.1127089374	22	6.779e−02	9.965e−01	COL1A1:148 HOXB4:151 COL2A1:163 APC:258 FBN1:5662.5 NCKAP1L:5662.5 MBTPS1:5662.5
MP0001845 abnormal inflammatory respons	0.0162193558	434	2.672e−01	9.965e−01	TRPV1:23 TNFSF10:36 IFNG:93 LRRK2:96 FOXP3:105 SERPINA6:139 IL12B:143
MP0004036 abnormal muscle relaxation	0.0084508145	30	8.731e−01	9.965e−01	MBNL2:223 MYL2:758 APOE:5662.5 TIMP3:5662.5 HIF1A:5662.5 SLC8A1:5662.5 ITGB1:5662.5
MP0002398 abnormal bone marrow	0.0145696917	651	2.357e−01	9.965e−01	TNFSF10:36 DNMT3B:57 CD86:62 FOXP3:105 NFKBIE:108 STAT2:129 IL12B:143
MP0000733 abnormal muscle development	0.0205007701	54	6.041e−01	9.965e−01	MGAT2:422 DLL3:631 MET:1007 CDKN1A:5662.5 IKBKB:5662.5 CTNNB1:5662.5 RELA:5662.5
MP0005423 abnormal somatic nervous	−0.0246959775	10	7.870e−01	9.965e−01	TAS1R1:1751 TAS1R3:1909 L1CAM:7003.5 SLC12A5:7003.5 EGR3:7003.5 DSCAM:7003.5 CNTN2:7003.5
MP0000230 abnormal systemic arterial	−0.0128745242	118	6.329e−01	9.965e−01	NOS3:244 PTGIS:303 TRPC6:316 SLC6A18:424 NOX1:627 IGHMBP2:1097 APEX1:1205
MP0003984 embryonic growth retardation	−0.0008023175	249	9.661e−01	9.965e−01	HUS1:54 ENPP2:110 KRT19:198 TSC1:209 LIG1:243 YBX1:306 CACNB2:335
MP0002733 abnormal thermal nociception	−0.0090805808	71	7.927e−01	9.965e−01	CLCN6:280 ADORA1:439 NTSR1:541 NTSR2:865 SEMA6C:1036 PTPRZ1:1062 OPRL1:1248
MP0005387 immune system phenotype	0.0247059831	89	4.244e−01	9.965e−01	FOXP3:105 GCNT1:310 RELB:378 ICAM1:477 IRF4:514 TLR9:517 CD79A:527
MP0005386 behavior/neurological phenoty	−0.0019891687	74	9.531e−01	9.965e−01	ARCN1:262 DICER1:391 WFS1:447 CCKAR:506 SCN8A:822 FXN:1088 DHCR7:1251
MP0005385 cardiovascular system phenoty	−0.0093601406	140	7.060e−01	9.965e−01	KCNH2:1 ADRB1:105 NOS3:244 CACNB2:335 RBP4:340 PTGIR:974 IGHMBP2:1097
MP0005384 cellular phenotype	−0.0113484904	341	4.862e−01	9.965e−01	DYNC2H1:2 TNS3:9 PMS2:15 CALR:51 HUS1:54 PLEC:83 ARID4B:88
MP0001545 abnormal hematopoietic system	−0.0144386933	29	7.884e−01	9.965e−01	LIG1:243 MMP9:328 MATK:904 RAD50:1004 EP400:1227 ABL1:1507 IFNGR1:7003.5

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
EBV.lymphocyte	0.031387813	576	1.322e−02	3.568e−01	CLEC17A:12 ANGPTL6:20 ATAD5:53 CD86:62 MIS18BP1:76 TCL1A:104 NFKBIE:108
testis	0.024870420	1196	7.854e−03	3.568e−01	SLC22A16:2 FAM169A:5 C1orf141:7 CATSPERE:8 RANBP17:16 RSPH14:21 C9orf135:22
amygdala	−0.206423696	7	5.868e−02	7.408e−01	PCDH15:18 PTPRZ1:1062 BCAN:1696 NEUROD6:2038 LHFPL3:7003.5 SLC38A8:7003.5 TNR:7003.5
nucleus.accumbens	−0.070729820	63	5.310e−02	7.408e−01	HTR1D:78 PENK:122 HTR6:125 INSYN2A:225 OTOF:292 DACT2:430 NECAB2:537
pancreas	0.051661524	81	1.097e−01	7.408e−01	FBXW12:1 CATSPERB:37 PRODH2:40 CELA3A:50 G6PC2:220 NPHS1:221 SERPINA4:278
peyers.patch	−0.041625224	152	7.936e−02	7.408e−01	SLC28A1:27 TRPM5:77 VIL1:236 ANPEP:308 TM4SF5:310 TRIM40:342 RNF186:484
tibial.nerve	−0.040507769	157	8.285e−02	7.408e−01	ITGB4:106 GNRH1:216 DCLK3:233 ITGA6:313 ENTPD2:323 ARHGAP39:442 AFAP1L2:458
transverse.colon	−0.042388836	128	1.004e−01	7.408e−01	DSG2:167 VIL1:236 TM4SF5:310 ATP10B:318 TRIM40:342 SELENBP1:403 ERN2:466
liver	0.025768373	294	1.358e−01	8.148e−01	AFP:6 AKR1D1:27 AFM:31 PRODH2:40 HSD17B2:52 ITIH4:67 SERPINA6:139
atrium.auricle	−0.038711792	78	2.395e−01	8.192e−01	MYBPHL:56 CHRNE:151 LMAN1L:264 MB:354 ADAMTS7:558 ADPRHL1:833 ALPK2:1002
blood	0.021266773	235	2.686e−01	8.192e−01	SHKBP1:26 R3HDM4:45 IFNG:93 TCL1A:104 GATA1:187 HCAR2:299 CXCR2:303
cerebellum	−0.021803810	286	2.129e−01	8.192e−01	APC2:25 SPHKAP:162 TRANK1:168 CNPY1:261 LMAN1L:264 POLE:278 MCF2L:317
ectocervix	−0.067268061	22	2.753e−01	8.192e−01	RIPOR3:770 F10:853 ADRA2A:1269 KCNK6:1350 COL27A1:1776 ESR1:2003 GDF7:7003.5
endocervix	−0.042400605	59	2.617e−01	8.192e−01	ZNF516:107 MLPH:152 NXNL2:240 KIF26A:621 TMEM132C:690 PWP2:788 CTSK:1080
esophagus.mucosa	−0.023530770	175	2.882e−01	8.192e−01	IQANK1:28 BNC1:44 EPS8L1:52 RNF222:79 FAM83H:117 N4BP3:176 STAP2:214
ovary	−0.034950263	81	2.793e−01	8.192e−01	TCF23:20 SUSD4:68 MMP21:87 GLI2:220 ADAMTS1:347 KLHDC8A:350 ADAMTS16:681
sigmoid.colon	−0.063546011	39	1.707e−01	8.192e−01	CHRNA3:45 TACR2:223 HMCN2:431 NTSR1:541 CSRP1:793 CACNA1H:1236 CHRM2:1921
skeletal.muscle	−0.027429239	190	1.977e−01	8.192e−01	SMTNL2:67 RORC:206 RGS9BP:246 ALPK3:325 MB:354 CHRNB1:388 TACC2:467
stomach	−0.045871998	74	1.744e−01	8.192e−01	MYH13:90 CAPN8:238 ATP4B:249 TM4SF5:310 CCKAR:506 CYP2S1:543 ATP4A:751
C1.spinal.cord	−0.025810898	134	3.063e−01	8.270e−01	ABCA2:35 ENPP2:110 ADARB2:242 S1PR5:307 PADI2:348 ACSBG1:368 MOG:392
breast	−0.073747509	12	3.767e−01	9.246e−01	TNN:351 THRSP:498 ALX4:636 ABCB5:7003.5 ADIPOQ:7003.5 CIDEC:7003.5 HCAR1:7003.5
cerebral.cortex	−0.029705985	78	3.667e−01	9.246e−01	CHRNB2:3 HTR3B:42 ADGRA1:74 KCNH3:95 KCNF1:374 ICAM5:437 FAM163B:566
aorta	−0.020851711	128	4.190e−01	9.379e−01	ADAMTSL3:140 PRDM16:270 PTGIS:303 HAPLN1:377 OGN:483 ITGA11:931 ST6GAL2:941
greater.omentum	−0.045178704	24	4.442e−01	9.379e−01	BNC1:44 SOX17:422 THRSP:498 ADAMTS9:1420 ALOX15:2285 FFAR3:7003.5 ADIPOQ:7003.5
left.ventricle	−0.031101851	57	4.183e−01	9.379e−01	MB:354 MYH7B:596 ADPRHL1:833 POPDC2:871 TNNT2:1423 MYOM2:1564 ASB11:1636
putamen	−0.066168687	10	4.690e−01	9.379e−01	RGS9:1595 OPALIN:1900 DIPK1C:2111 ANO3:7003.5 FAM237A:7003.5 GRM3:7003.5 SYNDIG1L:7003.5
suprapubic.skin	0.016628103	174	4.542e−01	9.379e−01	CD207:44 KRT16:168 CASP14:264 TFAP2C:295 IL1RL2:351 CAMSAP3:433 SLC10A6:441
adrenal.gland	−0.017280840	93	5.669e−01	9.626e−01	SLC47A1:298 RARRES2:359 TKFC:402 CELA1:441 CYB561A3:550 PLEK2:594 MAP3K15:624
bladder	0.005053095	18	9.409e−01	9.626e−01	ARL14:5662.5 CLEC3A:5662.5 CYP4B1:5662.5 KRT13:5662.5 PLA2G2F:5662.5 PLA2G4A:5662.5 SEMA3E:5662.5
Brodmann.area.24	0.004435267	21	9.440e−01	9.626e−01	DDN:685 NTRK2:836 CNTN5:5662.5 CRH:5662.5 DRD5:5662.5 FAM189A1:5662.5 GRIN3A:5662.5
Brodmann.area.9	−0.007610037	121	7.741e−01	9.626e−01	CHRNB2:3 HTR3B:42 CSRNP3:133 PCDHGC5:232 PCDH8:276 RNF157:433 DLGAP2:619
caudate.nucleus	−0.015812873	19	8.116e−01	9.626e−01	GPR37L1:1018 RLBP1:1213 PDE1B:1750 ANO3:7003.5 CHAT:7003.5 ETNPPL:7003.5 FAM237A:7003.5
cerebellar.hemisphere	−0.004975860	358	7.520e−01	9.626e−01	MINAR1:8 KIF26B:118 SPHKAP:162 CNPY1:261 LMAN1L:264 PDZD4:271 PER3:279
coronary.artery	−0.010150850	13	8.992e−01	9.626e−01	SUSD5:1542 ST6GALNAC5:1742 CCN3:7003.5 SPINK13:7003.5 MYO18B:7003.5 CCDC190:7003.5 CCL19:7003.5
cortex.kidney	−0.017927865	100	5.381e−01	9.626e−01	SLC28A1:27 SMIM32:89 SLC4A9:156 KLHDC7A:195 TM4SF5:310 RNF186:484 AQP6:532
esophagogastric.junction	0.054844763	5	6.711e−01	9.626e−01	ADCY5:5662.5 CHRM3:5662.5 COL4A5:5662.5 F2RL2:5662.5 GADL1:5662.5 NA NA
esophagus.muscularis.mucosa	0.016132862	18	8.129e−01	9.626e−01	JPH2:42 CCDC69:695 CHRM3:5662.5 F2RL2:5662.5 GADL1:5662.5 CYSLTR1:5662.5 KCNMB1:5662.5
fallopian.tube	−0.010225907	44	8.149e−01	9.626e−01	TCF23:20 HOXA7:696 ADGRG4:915 HOXA6:1369 CCDC17:1449 HOXC5:1553 NGF:1808
hippocampus.proper	0.059204914	6	6.156e−01	9.626e−01	CABP7:343 HRK:5662.5 GRP:5662.5 NEUROG3:5662.5 NTS:5662.5 NEUROD6:10628 NA
hypothalamus	0.002374136	45	9.562e−01	9.626e−01	GHRH:5662.5 NPY2R:5662.5 GRIN3A:5662.5 CSMD3:5662.5 INSM2:5662.5 PCDH19:5662.5 VSTM5:5662.5